Problems (please write down some middle steps to the final answers)

1. Distinguish between continuous and discontinuous variation in a population, and give some examples of each.
2. What are the central assumptions of the multifactorial hypothesis?
3. The table below shows a distribution of bristle number in a *Drosophila* population. Calculate the mean, variance, and standard deviation for these data.

Bristle number	Number of individuals
1	1
2	4
3	7
4	31
5	56
6	17
7	4

1.

Continuous variation: 表型无法分类。例：身高、体重。

Discontinuous variation:

Meristic traits: 表型可以通过数整数来记录。例：鸡下蛋的数量

Threshold traits: 多基因控制，基因数量达到阈值后显示表型。例：二型糖尿病。

2.

基因遵循孟德尔遗传定律，但可以以累积的方式对表型起作用。

基因的功能是加性的。

3. Mean：4.7

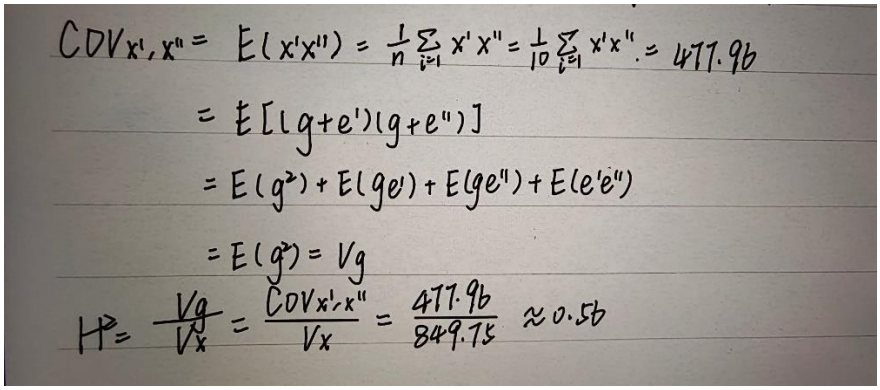
   Variance：1.11

   Standard deviation：1.05

Problems (please write down some middle steps to the final answers)

4. The table below contains measurements of total serum cholesterol (mg/dl) for 10 sets of monozygotic twins who were reared apart. Calculate the following: overall mean, overall variance, covariance between the twins, and broad-sense heritability (H^2).

X'	X''
228	222
186	152
204	220
142	185
226	210
217	190
207	226
185	213
179	159
170	129

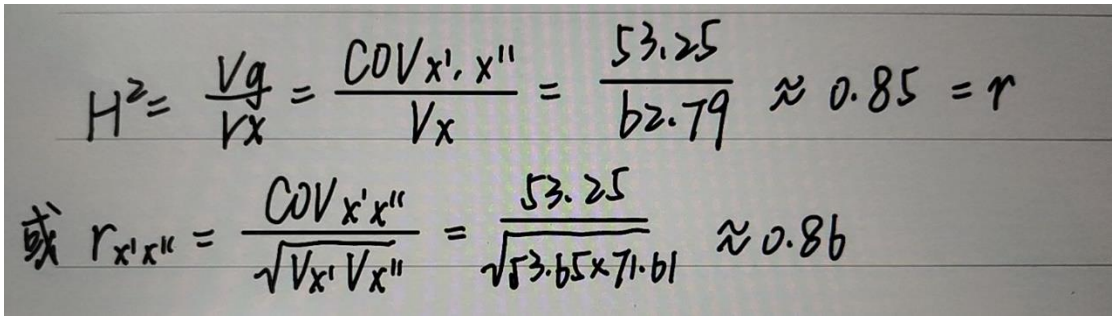


	X'	X''	overall
	228	222	
	186	152	
	204	220	
	142	185	
	226	210	
	217	190	
	207	226	
	185	213	
	179	159	
	170	129	
mean	194.4	190.6	192.5
variance	662.64	1029.64	849.75
covariance	477.96		
H^2	0.562471		

Problems (please write down some middle steps to the final answers)

5. The table below contains the height in centimeters for 10 sets of adult women twins. Calculate the correlation coefficient (r) between the heights of the sisters for the twin pairs.

Twin 1	Twin 2
158	163
156	150
172	173
156	154
160	163
159	153
170	174
177	174
165	168
172	165



	158	163
	156	150
	172	173
	156	154
	160	163
	159	153
	170	174
	177	174
	165	168
	172	165
variance	53.65	71.61
overall variance	62.79	
covariance	53.25	
H^2	0.85	
r X'X''	0.86	

Problems (please write down some middle steps to the final answers)

6. In a population of *Drosophila melanogaster* reared in the laboratory, the mean wing length is 0.55 mm and the range is 0.35 to 0.65. A geneticist selects a female with wings that are 0.42 mm in length and mates her with a male that has wings that are 0.56 mm in length.

   a. What is the expected wing length of their offspring if wing length has a narrow-sense heritability of 1.0?

   b. What is the expected wing length of their offspring if wing length has a narrow-sense heritability of 0.0?

